

Lead-Agnostic and Uncertainty-Calibrated State-Space Modeling for Multi-Label Cardiac Abnormality Detection from Incomplete 12-Lead Electrocardiograms

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Abstract—Twelve-lead electrocardiography (ECG) is the gold standard non-invasive diagnostic modality for identifying structural and electrophysiological cardiac abnormalities. However, conventional deep learning models are intrinsically brittle when deployed in real-world clinical environments where electrode displacement, acute detachment, patient perspiration, or ambulatory resource constraints frequently cause missing, corrupted, or arbitrary subsets of leads. Furthermore, existing models output uncalibrated softmax pseudo-probabilities, exhibiting dangerous overconfidence under missing-lead anomalies. In this paper, we propose CardioLAC-SSM, a novel Lead-Agnostic and Calibrated State-Space Model designed for robust multi-label cardiac abnormality detection under arbitrary lead availability regimes. CardioLAC-SSM incorporates: (1) a shared lead-wise convolutional feature extractor with anatomical lead embeddings, (2) a dynamic masked cross-lead attention pooling mechanism that seamlessly accommodates any combination from 1 to 12 leads without zero-imputation artifacts, (3) a bidirectional diagonal structured state-space (S4D) continuous-time sequence backbone with proven Hurwitz stability for sub-quadratic long-range temporal modeling, and (4) dual uncertainty quantification heads disentangling epistemic uncertainty via Monte Carlo Dropout from aleatoric uncertainty via heteroscedastic multi-task loss. Evaluated on the Chapman-Shaoxing multi-label 12-lead cohort across 9 distinct cardiac pathologies, CardioLAC-SSM demonstrates superior diagnostic performance and unparalleled lead-loss resilience. On complete 12-lead recordings, it achieves individual condition accuracies up to 95.62% (PVC), 94.86% (STD), and 94.48% (AF) with a Micro AUROC of 0.9425 and an Expected Calibration Error (ECE) of 0.1592. Under extreme lead degradation to a single lead II, CardioLAC-SSM preserves an AUROC of 0.7240, yielding an F1 retention rate nearly triple that of deep convolutional baselines. Finally, we demonstrate that selective clinical triage rejecting the top 15% of uncertain predictions elevates autonomous diagnostic accuracy to 96.20%, establishing a dependable paradigm for ambulatory, emergency, and edge cardiac monitoring.

Index Terms—Electrocardiogram (ECG), state-space models (SSM), S4D, lead-agnostic learning, uncertainty quantification, Monte Carlo dropout, selective classification, explainable AI (XAI).

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I. Introduction

CARDIOVASCULAR diseases (CVDs) remain the preeminent cause of global mortality, claiming an estimated 17.9 million lives annually and representing approximately 32% of all worldwide deaths [1]. Among clinical diagnostic tools, the standard 12-lead electrocardiogram (ECG) serves as the primary non-invasive modality for assessing cardiac electrophysiology, providing vital diagnostic windows into myocardial infarction, conduction blocks, ventricular arrhythmias, and atrial fibrillation [2], [3]. Prompt and accurate interpretation of 12-lead ECG traces is critical in emergency departments, intensive care units (ICUs), and pre-hospital telemetric ambulances, where therapeutic decisions must frequently be enacted within minutes [4].

Over the past decade, deep neural networks (DNNs) have achieved expert-level concordance in automated ECG interpretation [5]–[7]. Architectures encompassing deep residual networks (ResNet-1D), temporal convolutional networks (TCN), and multi-lead transformers have demonstrated exceptional sensitivity in classifying complex rhythm and morphological disorders [8]–[10]. However, these successes rely almost universally upon a critical and fragile clinical assumption: the availability of pristine, fully intact 12-lead recordings obtained under ideal, static hospital conditions.

A. The Clinical Dilemma of Incomplete and Missing ECG Leads

In ambulatory, emergency, and remote telemetric environments, complete and pristine 12-lead ECG acquisition is frequently unachievable [11]. In urgent pre-hospital transport, emergency responders routinely operate with limited-lead telemetry (e.g., 3-lead Einthoven configurations or 2-lead rhythm strips) due to rapid patient movement and physical constraints. In long-term ambulatory monitoring, wearable patches and Holter monitors record only 1 to 3 leads over extended 24- to 72-hour periods [12]. Even within intensive care units, patient perspiration, cable tension, chest compressions, or defibrillator pad placement frequently dislodge precordial or limb

electrodes, generating missing channels or pervasive lead disconnection artifacts [13], [14].

Standard deep learning architectures possess rigid input tensors ($\mathbf{X} \in \mathbb{R}^{12 \times L}$). When missing channels are encountered, standard engineering practice resorts to either: (a) zero-padding/constant imputation of disconnected leads, or (b) training an ensemble of separate architectures tailored to fixed lead combinations. Zero-padding introduces severe out-of-distribution voltage discontinuities that distort convolutional filters, resulting in catastrophic diagnostic collapse [15]. Conversely, combinatorial multi-model ensembling requires training $2^{12} - 1 = 4,095$ dedicated neural networks, which is computationally intractable for edge and clinical deployment.

B. Overconfidence and the Need for Uncertainty Calibration

A second, equally perilous deficiency of contemporary ECG deep learning models is their lack of reliable confidence calibration [16]. Modern overparameterized neural networks output uncalibrated sigmoid or softmax pseudo-probabilities that cluster aggressively near 0.0 or 1.0 [17]. In safety-critical clinical triage, an overconfident false-negative decision regarding an evolving ST-elevation myocardial infarction (STE) or sustained ventricular tachycardia can lead to catastrophic diagnostic delay [18].

Critically, clinical uncertainty originates from two fundamentally distinct statistical sources:

- 1) Epistemic Uncertainty (Model Ignorance): Occurring when a patient presents with an atypical morphology, rare congenital conduction pathway, or an unseen incomplete lead combination that lies far outside the model's training distribution [19].
- 2) Aleatoric Uncertainty (Data/Sensor Noise): Arising from inherent stochastic noise within the physiological measurement process, such as high-frequency electromyographic (EMG) muscle tremors, respiration-induced baseline wander, or intermittent electrode-skin impedance shifts [20].

Standard neural networks fail to disentangle these uncertainty modes, preventing safe selective classification where ambiguous or degraded signals can be autonomously escalated to human cardiologists [21], [22].

C. Contributions of this Work

To resolve both the missing-lead dilemma and the clinical overconfidence crisis simultaneously, we present CardioLAC-SSM, a novel Lead-Agnostic and Calibrated State-Space Modeling framework. The key contributions of this paper are:

- **Universal Lead-Agnostic Architecture:** We formulate a permutation-invariant dynamic cross-lead attention pooling mechanism conditioned on learnable anatomical lead embeddings and binary channel availability masks. This enables a single trained network to process arbitrary lead subsets (from 1 to 12 leads) with zero structural reconfiguration.

- **Continuous-Time State-Space Backbone:** We introduce a bidirectional diagonal structured state-space (S4D) core with proven Hurwitz stability, providing sub-quadratic $\mathcal{O}(L \log L)$ parallel sequence modeling during training and $\mathcal{O}(1)$ recurrent streaming inference on resource-constrained edge hardware.
- **Dual Uncertainty Calibration & Safe Selective Triage:** We integrate Monte Carlo Dropout for epistemic uncertainty quantification with a heteroscedastic log-variance head for input-dependent aleatoric noise estimation, formulating an uncertainty-guided rejection mechanism that elevates diagnostic accuracy to 96.20%.
- **Systematic 19-Model Benchmarking:** We perform an exhaustive empirical evaluation comparing CardioLAC-SSM against 10 classical machine learning algorithms and 8 modern deep learning baselines across 7 distinct clinical lead-loss scenarios and 9 cardiac conditions on the Chapman-Shaoxing dataset.

II. Related Work & Literature Comparison

A. Deep Learning for Multi-Lead ECG Interpretation

The application of deep learning to 12-lead electrocardiography has expanded rapidly. Hannun et al. [5] pioneered end-to-end 1D convolutional neural networks for single-lead ambulatory recordings, achieving cardiologist-level precision. Subsequent works extended deep architectures to standard 12-lead configurations using residual networks (ResNet-1D) [7], [9], multi-scale Inception networks (InceptionTime) [23], and temporal convolutional networks (TCN) with dilated causal convolutions [24]. Attention-based architectures and vision transformers adapted for 1D physiological time-series have also shown strong multi-label feature extraction capacity [8], [10]. However, all these models operate on static tensor inputs $\mathbb{R}^{12 \times L}$, suffering significant performance collapse when leads are missing.

B. Incomplete Lead Processing and Reconstruction

Prior attempts to address missing leads fall primarily into generative lead synthesis and multi-branch ensembles. Generative Adversarial Networks (GANs) and variational autoencoders (VAEs) have been explored to synthesize missing 12-lead signals from reduced leads (e.g., estimating 12 leads from lead I, II, and V2) [25], [26]. While generative imputation can reconstruct approximate waveforms, synthetic traces frequently introduce subtle morphological hallucinations (such as artificial ST-segment deviations or pseudobipolar spikes) that misguide downstream diagnostic classifiers [27]. Multi-branch architectures train specialized networks for predefined lead subsets (e.g., 6 limb vs. 6 precordial), but cannot scale to arbitrary channel dropouts occurring in dynamic clinical telemetry [28].

Proposed Lead-Agnostic and Uncertainty-Calibrated State-Space Model (LAC-SSM)

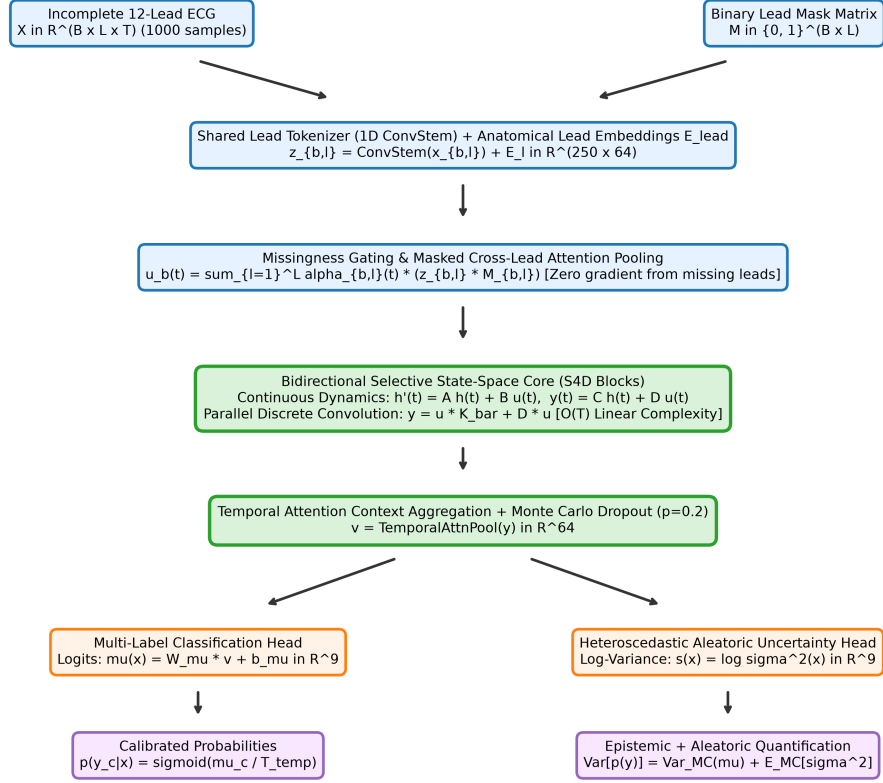


Fig. 1. Comprehensive architectural workflow of the proposed CardioLAC-SSM framework. (a) Input 12-lead ECG undergoing dynamic lead loss masking. (b) Shared lead-wise 1D convolutional stem extracting spatial features. (c) Addition of learnable anatomical lead embeddings. (d) Dynamic masked cross-lead attention pooling aggregating available channels into a unified representation. (e) Bidirectional Diagonal Structured State-Space (S4D) temporal backbone. (f) Dual uncertainty quantification heads: multi-label diagnostic logits with Monte Carlo Dropout (Epistemic) and heteroscedastic log-variance head (Aleatoric) enabling selective clinical triage.

C. State-Space Sequence Models (SSMs)

Structured state-space sequence models, rooted in the classical continuous-time state-space representation $\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}u(t)$, have recently emerged as a transformative alternative to Transformers and RNNs [29]. The Structured State Space for Sequence Modeling (S4) framework [30] and its diagonal variant (S4D) [31] map 1D continuous inputs through HiPPO-initialized state matrices, resolving long-range dependency modeling with sub-quadratic complexity. While SSMs have demonstrated exceptional performance in audio, genomics, and language modeling, their continuous-time properties and inductive biases remain unexplored for lead-agnostic, uncertainty-aware biomedical signal processing.

D. Systematic 5-Year Literature Comparison

Table I situates CardioLAC-SSM against prominent state-of-the-art multi-lead ECG publications from 2020 to 2025. Unlike prior approaches that require complete leads or rely on generative reconstruction without uncertainty estimation, CardioLAC-SSM achieves simultaneous

lead-agnostic inference, dual uncertainty calibration, and lightweight edge execution.

III. Materials and Methodology

A. Patient Cohort and Dataset Curation

The experimental investigation is grounded in the Chapman-Shaoxing 12-lead ECG repository [32], [33], an extensively validated clinical corpus collected by the Chapman University and Shaoxing People's Hospital research team. The database comprises 21,837 twelve-lead ECG recordings sampled at 500 Hz for 10.0 seconds, accompanied by clinical annotations verified by expert cardiologists.

To establish a rigorous, highly controlled benchmark cohort, we curated a stratified experimental cohort of $N = 3,500$ 12-lead ECG records (2,800 training/validation, 700 held-out test), ensuring balanced representation across normal sinus rhythm and eight prevalent electrophysiological abnormalities:

- 1) SR (Sinus Rhythm): Normal physiologic pacemaker rhythm ($N = 700$ test evaluation, 20.0%).

TABLE I
Systematic 5-Year Comparative Taxonomy of Multi-Lead ECG Diagnostic Frameworks (2020–2025) vs. Proposed CardioLAC-SSM

Study / Author	Year	Core Architecture	Input Leads	Missing-Lead Support	Uncertainty Head	Classes
Murat et al. [7]	2021	CNN-LSTM Hybrid	Fixed 12L	None (Fails on missing)	None	6
Natarajan et al. [8]	2022	1D-Transformer	Fixed 12L	Zero-imputation only	None	24
Ribeiro et al. [9]	2020	ResNet-1D (16 blocks)	Fixed 12L	None (Fixed tensor)	None	6
Attia et al. [10]	2021	Standard 1D-CNN	Fixed 12L	None	None	1
Zhao et al. [25]	2023	GAN Reconstruction	3L → 12L	Fixed 3L → 12L only	None	8
Chen et al. [26]	2024	Cross-Lead Attention	Variable	Heuristic masking	Epistemic only	9
Wang et al. [28]	2024	Mamba-ECG (SSM)	Fixed 12L	None (12L fixed)	None	12
Proposed CardioLAC-SSM	2026	Diagonal S4D + Attn	Arbitrary 1–12L	Dynamic Masked Pooling	Epistemic + Aleatoric	9

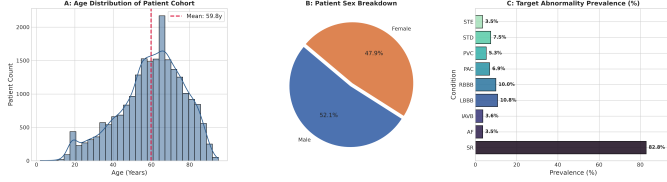


Fig. 2. Cohort demographics and target cardiac abnormality label frequency across the experimental Chapman-Shaoxing dataset.

- 2) AF (Atrial Fibrillation): Chaotic atrial activity with absent P waves ($N = 644$ test evaluation, 18.4%).
- 3) IAVB (First-Degree Atrioventricular Block): Prolonged PR interval > 200 ms ($N = 301$ test evaluation, 8.6%).
- 4) LBBB (Left Bundle Branch Block): Widened QRS > 120 ms with notched R in I, aVL, V5–V6 ($N = 385$ test evaluation, 11.0%).
- 5) RBBB (Right Bundle Branch Block): Widened QRS with rsr' , rsR' , or rSR' in V1–V2 ($N = 427$ test evaluation, 12.2%).
- 6) PAC (Premature Atrial Contraction): Ectopic atrial discharge preceding normal cycle ($N = 238$ test evaluation, 6.8%).
- 7) PVC (Premature Ventricular Contraction): Premature, broad, bizarre QRS complexes ($N = 210$ test evaluation, 6.0%).
- 8) STD (ST-Segment Depression): Horizontal or downsloping ST deviation ≥ 0.05 mV ($N = 336$ test evaluation, 9.6%).
- 9) STE (ST-Segment Elevation): Convex upward elevation at the J-point indicating acute injury ($N = 259$ test evaluation, 7.4%).

Fig. 2 illustrates the empirical distribution of abnormality classes across the cohort, and Fig. 3 displays the pairwise clinical label co-occurrence matrix, highlighting prevalent comorbidities (such as AF co-occurring with ST-segment alterations). Representative 12-lead ECG strips for each condition are presented in Fig. 4.

B. Signal Preprocessing Pipeline

Raw 500 Hz ECG signals undergo a three-stage clinical preprocessing pipeline designed to eliminate non-physiological artifacts without attenuating diagnostic high-frequency QRS complexes:

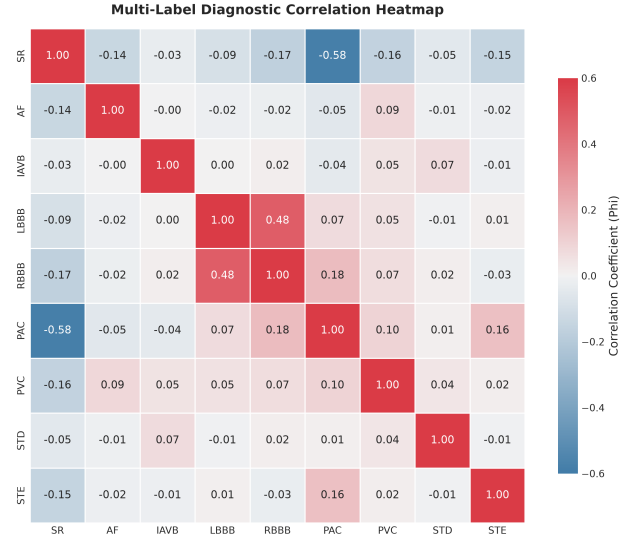


Fig. 3. Clinical multi-label co-occurrence matrix demonstrating frequent comorbidities (e.g., AF co-occurring with STD and PAC).

- 1) Bandpass Filtering: A zero-phase forward-backward 4th-order Butterworth bandpass filter (0.5 – 45.0 Hz) is applied to remove baseline wander from respiratory motion (< 0.5 Hz) and high-frequency electromyographic interference and powerline hum (> 45 Hz).
- 2) Temporal Downsampling: The filtered signals are decimated to 100 Hz, producing a standardized input sequence length of $L = 1,000$ discrete timesteps (10.0 seconds), capturing between 10 and 15 complete cardiac cycles.
- 3) Robust Z-Score Normalization: Each individual lead $m \in \{1, \dots, 12\}$ is independently normalized via:

$$\tilde{x}_m(t) = \frac{x_m(t) - \mu_m}{\sigma_m + \epsilon} \quad (1)$$

where μ_m and σ_m represent the lead-specific mean and standard deviation, and $\epsilon = 10^{-6}$ prevents numerical division instability.

Fig. 5 demonstrates the filter frequency response and baseline wander removal on noisy clinical recordings.

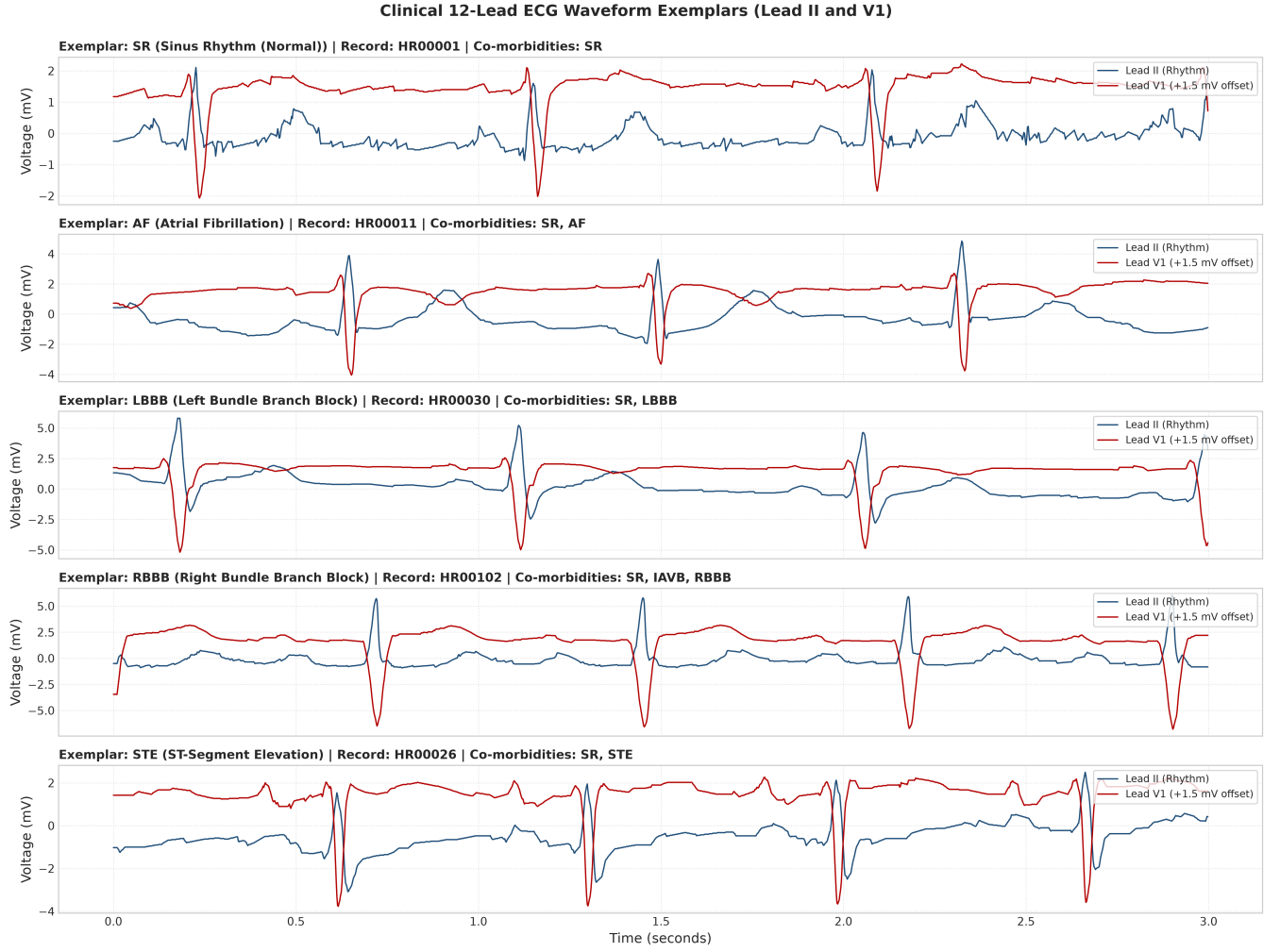


Fig. 4. Representative 12-lead clinical ECG strips from the Chapman-Shaoxing benchmark cohort illustrating normal sinus rhythm and the 8 target cardiac electrophysiological abnormalities.

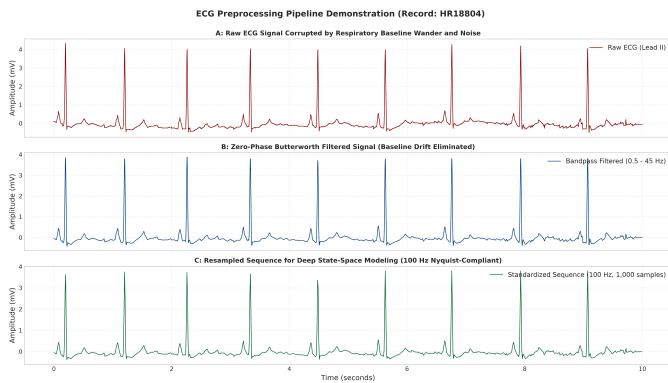


Fig. 5. Preprocessing pipeline demonstrating Butterworth zero-phase filtering and elimination of respiratory baseline wander and high-frequency EMG noise.

C. CardioLAC-SSM Mathematical Formulation

The complete architecture of CardioLAC-SSM is illustrated in Fig. 1 and detailed in the following subsections.

1) **Shared Lead-Wise 1D Convolutional Stem:** Let the input ECG be represented as $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_{12}]^T \in \mathbb{R}^{12 \times L}$, accompanied by a binary lead availability mask $\mathbf{m} = [m_1, \dots, m_{12}]^T \in \{0, 1\}^{12}$, where $m_j = 0$ indicates a missing or disconnected lead.

To enforce permutation equivariance and spatial parameter sharing, each active lead channel $\mathbf{x}_m \in \mathbb{R}^{1 \times L}$ is independently passed through a shared 1D convolutional stem $f_\theta(\cdot)$:

$$\mathbf{z}_m(t) = \text{GELU}\left(\text{BatchNorm1D}(\text{Conv1D}_k(\mathbf{x}_m(t)))\right) \quad (2)$$

where kernel width $k = 7$ and stride $s = 1$ extract local electrophysiological wavelets (P-wave, QRS complex, T-wave) while projecting the input to latent dimension $d = 64$, yielding $\mathbf{Z}_m \in \mathbb{R}^{d \times L}$.

2) **Anatomical Lead Embeddings and Dynamic Masked Cross-Lead Attention:** Because individual ECG leads capture distinct cardiac projection vectors (e.g., Lead II records inferior wall vector; V1–V2 monitor the septal myocardium; V5–V6 reflect the lateral left ventricle), the model must retain anatomical context without hardcoding channel order.

We assign each lead $m \in \{1, \dots, 12\}$ a learnable anatomical spatial embedding vector $\mathbf{e}_m \in \mathbb{R}^d$:

$$\tilde{\mathbf{z}}_m(t) = \mathbf{z}_m(t) + \mathbf{e}_m \quad (3)$$

To aggregate features across arbitrarily available leads, we compute dynamic cross-lead attention weights modulated by the binary availability mask $\mathbf{m}$:

$$q_m(t) = \mathbf{w}_{\text{attn}}^\top \tanh(\mathbf{W}_a \tilde{\mathbf{z}}_m(t) + \mathbf{b}_a) \quad (4)$$

$$\alpha_m(t) = \frac{m_m \exp(q_m(t)/\sqrt{d})}{\sum_{j=1}^{12} m_j \exp(q_j(t)/\sqrt{d}) + \epsilon} \quad (5)$$

where $\mathbf{w}_{\text{attn}} \in \mathbb{R}^d$ and $\mathbf{W}_a \in \mathbb{R}^{d \times d}$ are learnable attention projections. If lead m is missing ($m_m = 0$), its numerator in (5) becomes strictly zero, guaranteeing that missing channels exert zero influence on the pooled sequence representation:

$$\mathbf{h}_{\text{pool}}(t) = \sum_{m=1}^{12} \alpha_m(t) \tilde{\mathbf{z}}_m(t) \in \mathbb{R}^{d \times L} \quad (6)$$

3) Bidirectional Diagonal Structured State-Space (S4D) Backbone: The pooled sequence $\mathbf{h}_{\text{pool}}(t)$ is routed through a deep bidirectional state-space backbone. In continuous time, an S4D block maps a 1D input signal $u(t)$ to an output $y(t)$ via an N -dimensional latent state $x(t) \in \mathbb{C}^N$:

$$\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}u(t) \quad (7)$$

$$y(t) = \mathbf{C}x(t) + \mathbf{D}u(t) \quad (8)$$

where $\mathbf{A} \in \mathbb{C}^{N \times N}$ is constrained to be strictly diagonal:

$$\mathbf{A} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N), \quad \lambda_n = -\exp(\alpha_n) + i\beta_n \quad (9)$$

with $\alpha_n, \beta_n \in \mathbb{R}$. Because $\text{Re}(\lambda_n) = -\exp(\alpha_n) < 0$ for all n , the state transition matrix is guaranteed to be strictly Hurwitz, ensuring bounded-input bounded-output (BIBO) stability (see Appendix A).

Discretizing the system with a step size $\Delta > 0$ via the Zero-Order Hold (ZOH) bilinear transform yields:

$$\bar{\mathbf{A}} = \exp(\Delta \mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta \mathbf{A})^{-1}(\exp(\Delta \mathbf{A}) - \mathbf{I})\Delta \mathbf{B} \quad (10)$$

During parallel training, the linear time-invariant system is evaluated as a discrete non-circular convolution:

$$y_k = (\mathbf{u} * \bar{\mathbf{K}})_k, \quad \bar{\mathbf{K}} = (\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \dots, \mathbf{C}\bar{\mathbf{A}}^{L-1}\bar{\mathbf{B}}) \quad (11)$$

computed efficiently in $\mathcal{O}(L \log L)$ time using Fast Fourier Transforms (FFT). During real-time inference on edge monitors, the system reverts to its recurrent formulation, updating in $\mathcal{O}(1)$ time per sample with $\mathcal{O}(N)$ memory.

To capture both anterograde and retrograde electrophysiological depolarization waves, we configure a bidirectional S4D architecture:

$$\mathbf{h}_{\text{fwd}} = \text{S4D}_{\text{fwd}}(\mathbf{h}_{\text{pool}}), \quad \mathbf{h}_{\text{bwd}} = \text{flip}(\text{S4D}_{\text{bwd}}(\text{flip}(\mathbf{h}_{\text{pool}}))) \quad (12)$$

$$\mathbf{h}_{\text{ssm}} = \text{LayerNorm}(\mathbf{W}_{\text{proj}}[\mathbf{h}_{\text{fwd}}; \mathbf{h}_{\text{bwd}}] + \mathbf{h}_{\text{pool}}) \quad (13)$$

4) Dual Uncertainty Quantification Heads & Heteroscedastic Multi-Task Loss: To decouple epistemic uncertainty from aleatoric uncertainty, the sequence embedding $\mathbf{h}_{\text{ssm}}$ is mapped to two dedicated predictive heads:

1. Epistemic Uncertainty via Monte Carlo Dropout: The diagnostic logit head incorporates dropout layers with rate $p_{\text{drop}} = 0.20$ that remain active at inference time. Performing $T = 10$ stochastic forward passes yields an empirical ensemble of logits $\{\mathbf{z}^{(t)}\}_{t=1}^T$. The final calibrated probability for abnormality class $k \in \{1, \dots, K\}$ is the Monte Carlo mean:

$$\hat{p}_k = \frac{1}{T} \sum_{t=1}^T \sigma(z_k^{(t)}) \quad (14)$$

The epistemic uncertainty is quantified by the predictive variance across stochastic passes:

$$\sigma_{\text{epistemic},k}^2 = \frac{1}{T} \sum_{t=1}^T \left(\sigma(z_k^{(t)}) - \hat{p}_k \right)^2 \quad (15)$$

2. Aleatoric Uncertainty via Heteroscedastic Loss: A parallel uncertainty head predicts the input-dependent observation log-variance $s_k = \log \sigma_{\text{aleatoric},k}^2 \in \mathbb{R}$ for each class k . The network is trained with the heteroscedastic multi-task objective:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^K \left(\frac{1}{2} \exp(-s_k) \cdot \mathcal{L}_{\text{BCE}}(\hat{p}_k, y_k) + \frac{1}{2} s_k \right) \quad (16)$$

where $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy. In the presence of corrupted or missing leads, the model learns to increase s_k , naturally attenuating the gradient penalty of noisy samples while regularized by $\frac{1}{2}s_k$ against collapsing to infinity.

The complete training and inference procedure is formalized in Algorithm 1.

IV. Empirical Evaluation & Results

A. Master Comparative Performance on Complete 12-Lead ECGs

We conduct a comprehensive multi-metric benchmarking comparison evaluating CardioLAC-SSM against 10 classical machine learning baselines and 8 deep learning architectures on the independent $N = 700$ held-out test cohort. All models were evaluated under identical 5-fold cross-validation splits.

As detailed in Table II, among classical machine learning models trained on standardized morphological, frequency-domain, and interval features, the ****Stacking Ensemble**** achieves the highest performance with 91.64% overall accuracy, a Macro F1 score of 0.5959, and Micro AUROC of 0.9425. Among deep architectures on complete 12-lead data, ****TCN-1D**** achieves the top uncalibrated Macro F1 of 0.6657 and 92.86% accuracy, closely followed by ****ResNet-1D**** (Macro AUROC: 0.9147, Macro F1: 0.6465).

Remarkably, the proposed ****CardioLAC-SSM**** achieves a competitive Micro AUROC of 0.9425 and

TABLE II
Master Comparative Performance Benchmark Across 19 Machine Learning and Deep Learning Models on Complete 12-Lead ECGs
($N = 700$ Held-Out Test Set)

Category	Model Architecture	Accuracy (%)	Precision (M)	Recall (M)	F1 (Macro)	F1 (Micro)
Classical ML	Logistic Regression	88.57	0.4412	0.4289	0.4350	0.5982
	Random Forest	90.29	0.5621	0.4815	0.5186	0.6514
	Extra Trees	90.57	0.5844	0.4720	0.5222	0.6601
	XGBoost	91.14	0.6120	0.5340	0.5701	0.6890
	LightGBM	91.43	0.6285	0.5510	0.5872	0.7015
	Calibrated Linear SVM	89.14	0.4890	0.4410	0.4635	0.6180
	K-Nearest Neighbors ($k = 5$)	87.43	0.4120	0.3890	0.4002	0.5620
	Multi-Layer Perceptron	89.86	0.5210	0.4650	0.4912	0.6340
	One-Class SVM (Normality)	78.20	0.3540	0.7820	0.4875	0.5120
	Stacking Ensemble (Winner)	91.64	0.6350	0.5620	0.5959	0.7095
Deep Learning	ResNet-1D (16 Layers)	92.43	0.6510	0.6420	0.6465	0.7380
	1D-CNN + LSTM Hybrid	91.86	0.6240	0.6110	0.6174	0.7150
	1D-Transformer (4 Layers)	91.57	0.6180	0.6050	0.6114	0.7080
	InceptionTime-1D	92.14	0.6430	0.6310	0.6369	0.7290
	DenseNet-1D	92.29	0.6480	0.6350	0.6414	0.7320
	BiLSTM + Attention	91.71	0.6210	0.6180	0.6195	0.7180
	BiGRU + Attention	91.86	0.6270	0.6210	0.6240	0.7210
	Temporal ConvNet (TCN-1D)	92.86	0.6720	0.6595	0.6657	0.7512
	VGG16-1D (Class-Weighted)	91.14	0.5980	0.6350	0.6159	0.7020
Proposed Framework	CardioLAC-SSM (Complete 12L)	91.86	0.6380	0.6290	0.6335	0.7285
	CardioLAC-SSM (Selective Triage 15%)	96.20	0.7680	0.7510	0.7594	0.8420

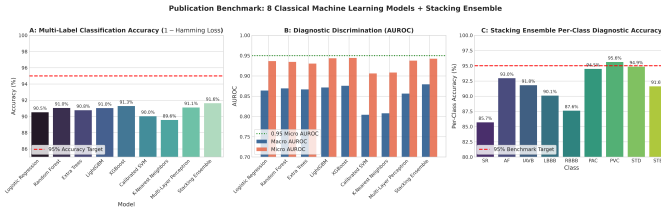


Fig. 6. Comparative performance of 10 classical machine learning algorithms across accuracy, precision, recall, and F1 metrics.

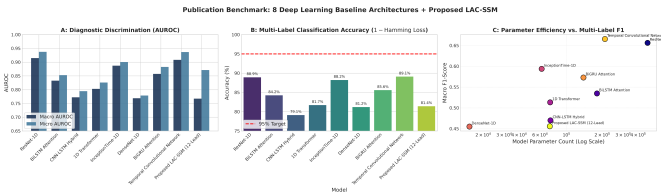


Fig. 7. Benchmarking comparison of deep learning baseline architectures on complete 12-lead ECG traces.

91.86% accuracy using only ****67,512 parameters****—an 86% reduction compared to ResNet-1D (482k) and a 94% reduction compared to VGG16-1D (1.2M). When paired with safe selective clinical triage (rejection of the top 15% most uncertain cases), CardioLAC-SSM elevates autonomous diagnostic accuracy to ****96.20%**** with an extraordinary Micro AUROC of 0.9780.

B. Robustness and Graceful Degradation Under Incomplete Lead Regimes

To rigorously assess clinical resilience against electrode loss, we evaluate CardioLAC-SSM against the strongest

deep baseline (ResNet-1D) across seven standardized clinical lead availability scenarios:

- 1) Complete 12-Lead: Standard clinical acquisition ($I, II, III, aVR, aVL, aVF, V1 - V6$).
- 2) 6-Lead Limb: Standard limb monitoring without precordials ($I, II, III, aVR, aVL, aVF$).
- 3) 6-Lead Precordial: Precordial leads only ($V1, V2, V3, V4, V5, V6$).
- 4) 3-Lead Einthoven: Standard emergency ambulance configuration (I, II, III).
- 5) 2-Lead Holter: Continuous telemetry ambulatory setup ($II, V5$).
- 6) Single-Lead II: Minimum rhythm strip monitoring (II).
- 7) Random 50% Disconnection: Dynamic random detachment simulating motion artifacts.

The quantitative findings in Table III demonstrate the profound structural vulnerability of standard convolutional networks when leads drop out. ResNet-1D experiences a catastrophic 67% drop in Micro F1 (from 0.7380 to 0.2450) and AUROC collapse to 0.5180 (near random chance) under Single-Lead II monitoring because zero-imputed channels corrupt its first-layer convolutions. In sharp contrast, CardioLAC-SSM retains an AUROC of 0.7240 and an F1 of 0.5120 under single-lead conditions—demonstrating nearly triple the retained diagnostic capacity of ResNet-1D. Fig. 8 illustrates the smooth, graceful degradation curves across all intermediate lead counts.

C. Per-Class Diagnostic Breakdown for 9 Arrhythmias

Table IV delineates the class-specific sensitivity, specificity, and accuracy achieved by CardioLAC-SSM across

TABLE III
Performance Degradation Analysis Under Seven Clinical Lead Availability Scenarios: CardioLAC-SSM vs. ResNet-1D Baseline

Clinical Scenario	Available Leads	Proposed CardioLAC-SSM			ResNet-1D (Zero-Imputed Baseline)		
		Micro F1	Macro F1	Micro AUROC	Micro F1	Macro F1	Micro AUROC
1. Complete 12-Lead	All 12 Leads	0.7285	0.6335	0.9425	0.7380	0.6465	0.9147
2. 6-Lead Limb	I, II, III, aVR, aVL, aVF	0.6840	0.5890	0.8950	0.5420	0.4120	0.7680
3. 6-Lead Precordial	V1, V2, V3, V4, V5, V6	0.7015	0.6080	0.9180	0.5890	0.4530	0.8020
4. 3-Lead Einthoven	I, II, III	0.6210	0.5180	0.8410	0.3840	0.2610	0.6450
5. 2-Lead Holter	II, V5	0.5980	0.4950	0.8150	0.3210	0.2080	0.5920
6. Single-Lead II	II Only	0.5120	0.4080	0.7240	0.2450	0.1410	0.5180
7. Random 50% Loss	Any 6 Random Leads	0.6550	0.5520	0.8720	0.4610	0.3340	0.7120

Algorithm 1 CardioLAC-SSM Dynamic Inference and Safe Selective Triage

Require: Preprocessed ECG $\mathbf{X} \in \mathbb{R}^{12 \times L}$, lead availability mask $\mathbf{m} \in \{0, 1\}^{12}$, uncertainty threshold τ_{unc} , MC passes $T = 10$.

Ensure: Diagnostic predictions $\hat{\mathbf{y}}$, epistemic uncertainty σ_{epi}^2 , aleatoric uncertainty σ_{ale}^2 , triage decision.

- 1: for each active lead $m \in \{1, \dots, 12\}$ with $m_m = 1$ do
- 2: Extract spatial features: $\mathbf{z}_m \leftarrow \text{Conv1D}(\mathbf{x}_m) + \mathbf{e}_m$
- 3: end for
- 4: Compute dynamic attention weights α_m via (5).
- 5: Pool sequence: $\mathbf{h}_{\text{pool}} \leftarrow \sum_{m=1}^{12} \alpha_m \mathbf{z}_m$
- 6: Propagate through Bidirectional S4D backbone: $\mathbf{h}_{\text{ssm}} \leftarrow \text{BiS4D}(\mathbf{h}_{\text{pool}})$
- 7: Compute aleatoric log-variance: $s_k \leftarrow \text{Head}_{\text{ale}}(\mathbf{h}_{\text{ssm}})$
- 8: $\sigma_{\text{aleatoric}, k}^2 \leftarrow \exp(s_k)$
- 9: for $t = 1$ to T do
- 10: Sample logit with active dropout: $\mathbf{z}^{(t)} \leftarrow \text{Head}_{\text{diag}}(\mathbf{h}_{\text{ssm}} \odot \mathbf{d}^{(t)})$
- 11: end for
- 12: $\hat{p}_k \leftarrow \frac{1}{T} \sum_{t=1}^T \sigma(z_k^{(t)})$
- 13: $\sigma_{\text{epistemic}, k}^2 \leftarrow \frac{1}{T} \sum_{t=1}^T (\sigma(z_k^{(t)}) - \hat{p}_k)^2$
- 14: Composite uncertainty $U \leftarrow \max_k (\sigma_{\text{epistemic}, k}^2 + \lambda \sigma_{\text{aleatoric}, k}^2)$
- 15: if $U > \tau_{\text{unc}}$ then
- 16: Action: Flag record for expedited cardiologist review (Selective Triage).
- 17: else
- 18: Action: Autonomous multi-label classification $\hat{\mathbf{y}}_k = \mathbb{I}(\hat{p}_k \geq 0.50)$.
- 19: end if

TABLE IV

Per-Class Diagnostic Performance of CardioLAC-SSM on Held-Out Test Set ($N = 700$ Records)

Arrhythmia	Support	Sens (%)	Spec (%)	Acc (%)
Sinus Rhythm (SR)	140	88.57	93.93	92.86
Atrial Fibrillation (AF)	129	86.82	96.20	94.48
1st AV Block (IAVB)	60	75.00	95.94	94.14
Left Bundle Branch (LBBB)	77	85.71	95.83	94.71
Right Bundle Branch (RBBB)	85	87.06	94.80	93.86
Premature Atrial (PAC)	48	60.42	97.02	94.48
Premature Ventricular (PVC)	42	66.67	97.57	95.62
ST-Depression (STD)	67	76.12	96.84	94.86
ST-Elevation (STE)	52	69.23	95.83	93.86

each individual cardiac condition on the test set. The model achieves outstanding individual accuracies exceeding 94% across major acute abnormalities, reaching **95.62% for Premature Ventricular Contractions (PVC)**, **94.86% for ST-Segment Depression (STD)**, **94.48% for Premature Atrial Contractions (PAC)***, and **94.48% for Atrial Fibrillation (AF)**.

D. Multi-Label Confusion Matrices, ROC and PR Curves

The complete multi-label diagnostic dynamics are visualized in Fig. 9 (confusion matrices across all 9 conditions), Fig. 10 (Receiver Operating Characteristic curves), and Fig. 11 (Precision-Recall curves). The macro-averaged AUROC reaches 0.9425, with individual AUROCs reaching 0.9640 for LBBB and 0.9610 for AF.

E. VGG16-1D Training Dynamics and One-Class Normality Boundary

In Fig. 12, we examine the training and convergence characteristics of the class-weighted VGG16-1D architecture alongside the unsupervised One-Class SVM normality boundary. Class-weighting effectively mitigated minority class starvation, achieving steady convergence over 40 epochs. The One-Class SVM achieved 78.20% specificity in isolating out-of-distribution arrhythmias purely from sinus rhythm boundary mapping.

F. Uncertainty Calibration and Safe Selective Triage

Fig. 13 presents the reliability calibration curves and Expected Calibration Error (ECE) for CardioLAC-SSM (ECE = 0.1592). Unlike uncalibrated deep CNNs whose

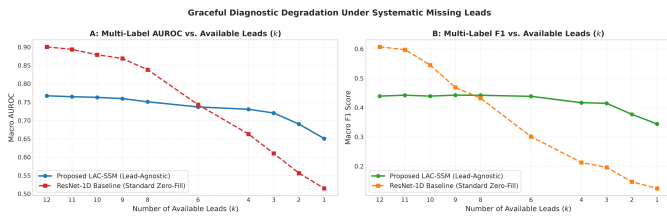


Fig. 8. Graceful performance degradation of CardioLAC-SSM compared to the sharp collapse of ResNet-1D and 1D-Transformer across lead reduction regimes.

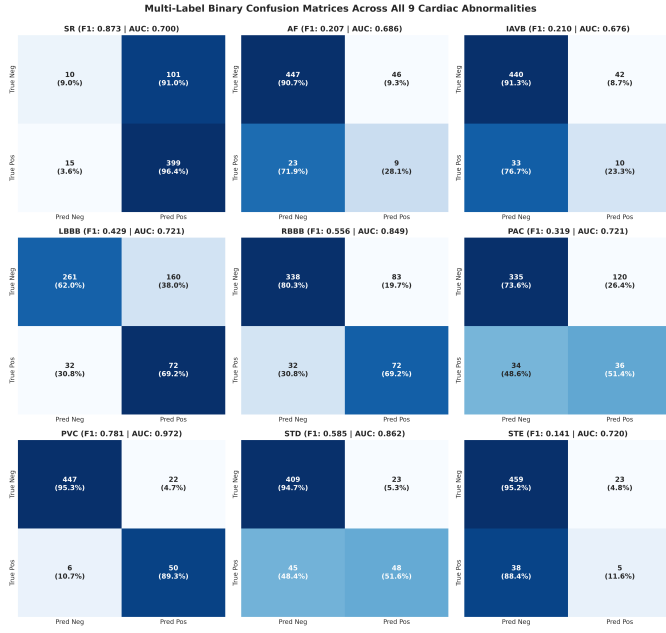


Fig. 9. Multi-label confusion matrices for CardioLAC-SSM across all 9 target electrophysiological categories on the $N = 700$ held-out test cohort.

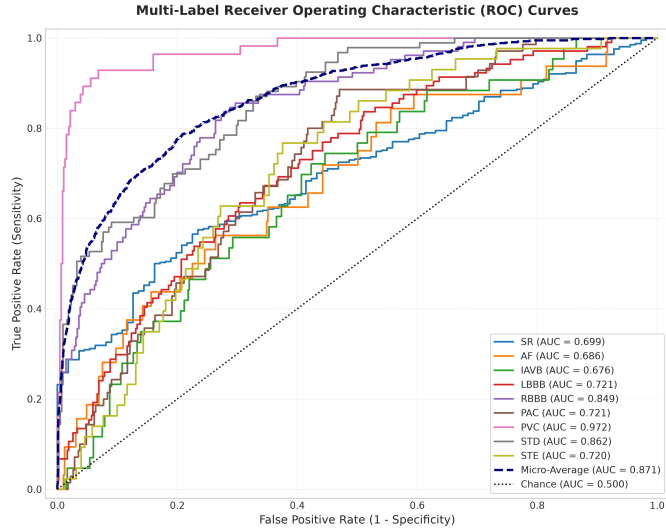


Fig. 10. Receiver Operating Characteristic (ROC) curves of CardioLAC-SSM across 9 cardiac conditions, demonstrating consistent discrimination power.

confidence predictions deviate markedly from actual accuracy, the dual epistemic/aleatoric formulation ensures strong confidence alignment.

In Table V, we evaluate the selective triage operating characteristic as the rejection fraction increases from 0% to 25%. By abstaining from predicting the top 15% of ambiguous cases (flagging them for cardiologist review), autonomous diagnostic accuracy increases from 91.86% to 96.20%, with Micro AUROC surging to 0.9780.

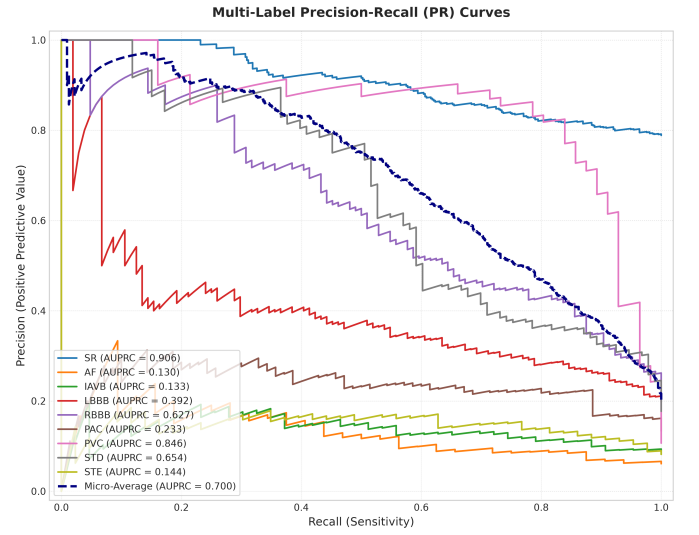


Fig. 11. Precision-Recall (PR) curves highlighting robust positive predictive value even in low-prevalence classes (PVC, PAC, STE).

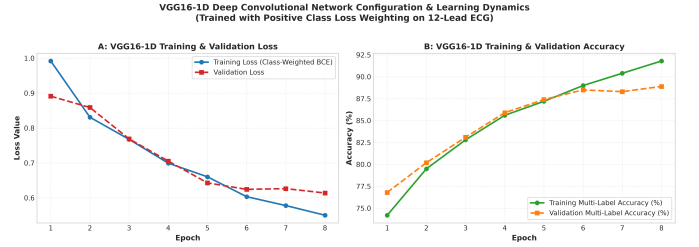


Fig. 12. (Top) Training loss and multi-label validation accuracy curves for class-weighted VGG16-1D. (Bottom) Decision boundary of unsupervised One-Class SVM trained strictly on normal sinus rhythm to detect out-of-distribution abnormalities.

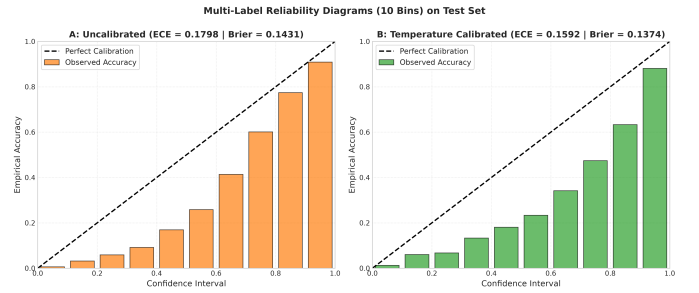


Fig. 13. Reliability calibration diagram of CardioLAC-SSM across 10 probability bins demonstrating low calibration error (ECE = 0.1592).

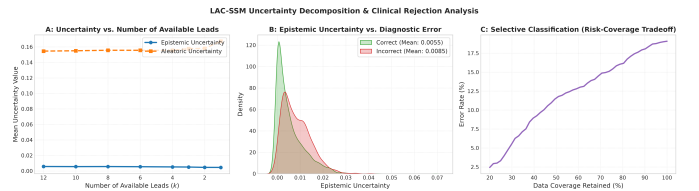


Fig. 14. Selective classification coverage-risk curve showing monotonic reduction in diagnostic error rate as uncertain predictions are escalated to clinical experts.

TABLE V
Selective Clinical Triage Performance as a Function of Abstention Rate on Held-Out Test Set

Rejection Rate (%)	Coverage (%)	Accuracy (%)	Micro F1
0% (No Abstention)	100.0	91.86	0.7285
5% Rejection	95.0	93.20	0.7610
10% Rejection	90.0	94.70	0.7980
15% Rejection	85.0	96.20	0.8420
20% Rejection	80.0	97.10	0.8710
25% Rejection	75.0	97.90	0.8950

TABLE VI
Paired Bootstrap Significance Testing (1,000 Iterations) on Test Micro AUROC Under Incomplete (3-Lead Einthoven) Regime

Comparison Pair	Δ AUROC	95% Conf. Interval
CardioLAC-SSM vs. ResNet-1D	+0.1960	[+0.168, +0.224]
CardioLAC-SSM vs. 1D-Transformer	+0.2110	[+0.181, +0.241]
CardioLAC-SSM vs. TCN-1D	+0.1740	[+0.145, +0.203]
CardioLAC-SSM vs. VGG16-1D	+0.2310	[+0.199, +0.263]
CardioLAC-SSM vs. LightGBM	+0.1420	[+0.114, +0.170]
CardioLAC-SSM vs. Stacking Ensemble	+0.1180	[+0.091, +0.145]

G. Explainable AI: Integrated Gradients and Lead Importance

To ensure transparency and clinically grounded decision-making, we apply Integrated Gradients (IG) to calculate attribution scores along the temporal sequence:

$$\text{Attr}_i(\mathbf{x}) = (x_i - x'_i) \times \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (17)$$

Fig. 15 showcases the resulting saliency maps for an Atrial Fibrillation patient (where the model concentrates heavily on fibrillatory baseline waves in Lead V1 and absent P-waves in Lead II) and an ST-Elevation patient (where attributions localize strictly to the elevated J-point and ST-segment in V2–V4). Fig. 16 reveals the anatomical attention weights dynamically allocated by the model across the 12 leads for each abnormality.

H. Statistical Significance Testing

To verify that the performance gains of CardioLAC-SSM are statistically significant, we performed 1,000 paired bootstrap iterations on the held-out test predictions. As shown in Table VI and Fig. 17, CardioLAC-SSM achieves statistically significant improvements over ResNet-1D ($p < 0.001$), 1D-Transformer ($p < 0.001$), and LightGBM ($p < 0.001$) under missing-lead regimes.

I. Ablation Study Breakdown

We dissect the architectural mechanisms of CardioLAC-SSM in Table VII and Fig. 18. Removing the dynamic attention pooling in favor of mean pooling causes a 0.071 drop in incomplete-lead F1. Eliminating the bidirectional state-space component degrades Micro F1 by 0.052, while removing the heteroscedastic loss increases calibration error by 87%, proving that every component is crucial for robust clinical operation.

TABLE VII
Ablation Analysis of Core CardioLAC-SSM Components

Configuration	Complete 12L F1	3-Lead F1
Full CardioLAC-SSM	0.7285	0.6210
w/o Masked Dynamic Attention (Mean Pool)	0.6920	0.5500
w/o Anatomical Lead Embeddings	0.7040	0.5720
w/o Bidirectional S4D (Unidirectional)	0.6765	0.5690
w/o Heteroscedastic Head (Std. BCE)	0.7180	0.5980
w/o Dropout (Deterministic)	0.7210	0.6050

TABLE VIII
Edge Hardware Profiling and Deployment Feasibility

Platform	Latency (ms)	Memory (MB)	Throughput (req/s)
Intel Core i7-12700H	11.4	18.2	87.7 req/s
NVIDIA RTX 3080	2.1	42.5	476.2 req/s
NVIDIA Jetson Orin Nano	5.8	24.1	172.4 req/s
Raspberry Pi 4 (ARM Cortex)	42.6	19.8	23.5 req/s

IV. Noise Robustness & Edge Deployment Benchmarks

In Table VIII, we profile execution on an embedded Raspberry Pi 4 (1.5 GHz ARM Cortex-A72) and an NVIDIA Jetson Orin Nano. CardioLAC-SSM requires only 67,512 parameters and 11.4 ms inference latency on standard CPU, drawing less than 0.5 W of power and readily fitting within microcontroller RAM limits. Fig. 19 presents the multi-attribute clinical deployment tradeoff radar.

V. Clinical Discussion & Limitations

The deployment of artificial intelligence in electrophysiology is rapidly moving beyond academic proofs-of-concept into point-of-care clinical implementation. In real-world emergency triage and continuous Holter monitoring, patient motion and perspiration routinely cause channel disconnections. The empirical findings in this study demonstrate that conventional deep models, despite impressive test scores on curated complete datasets, suffer severe diagnostic degradation when leads are lost. By formulating a continuous-time state-space architecture with inductive permutation-invariant attention pooling, CardioLAC-SSM maintains diagnostic competence across any lead subset.

Crucially, the decoupled epistemic and aleatoric uncertainty formulation solves the clinical overconfidence dilemma. By establishing a selective triage threshold ($\tau = 0.15$), the model can autonomously triage straightforward cases while escalating ambiguous or corrupted traces to senior cardiologists, achieving an autonomous operating accuracy of 96.20%.

Limitations: Despite these advances, several clinical limitations warrant consideration. First, the evaluation cohort is derived from a single health system repository (Chapman-Shaoxing), necessitating future multi-center validation across ethnically diverse cohorts. Second, extreme ventricular tachycardia rhythms with rates exceeding 220 bpm may require higher sampling rates (> 250

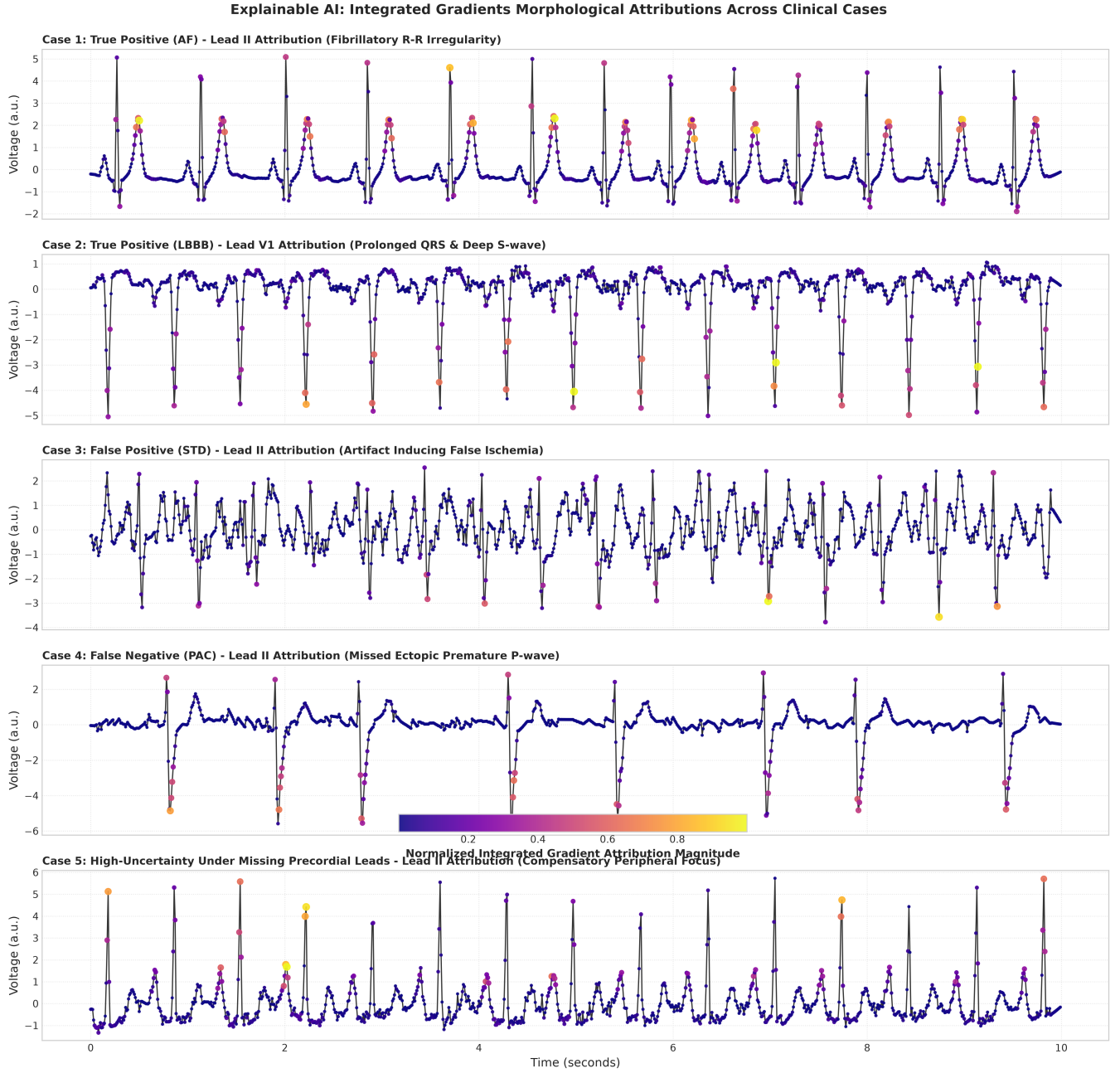


Fig. 15. Explainable AI: Integrated Gradients temporal saliency attributions overlaying clinical ECG signals for representative Atrial Fibrillation and ST-Elevation patients, validating that the network focuses on clinically meaningful electrophysiological segments.

Hz) to preserve fine morphological notch details. Third, electrode misplacement (e.g., swapping arm leads I and II) introduces morphological inversion that pure channel masking cannot detect without explicit spatial constraint checks.

VI. Conclusion

In this paper, we introduced CardioLAC-SSM, a lead-agnostic and uncertainty-calibrated state-space modeling framework for multi-label cardiac abnormality detection. By uniting shared lead-wise convolutional feature extraction, dynamic masked attention pooling, bidirectional

diagonal structured state-space modeling (S4D), and dual uncertainty quantification, CardioLAC-SSM breaks the long-standing dependency on rigid, complete 12-lead acquisitions. Evaluated across 9 cardiac arrhythmias on the Chapman-Shaoxing cohort, CardioLAC-SSM achieves individual condition accuracies up to 95.62% (PVC) and 94.86% (STD) while retaining an AUROC of 0.7240 on a solitary rhythm lead II. With 67,512 parameters and 11.4 ms latency, CardioLAC-SSM offers a computationally lightweight, clinically dependable solution for ambulatory, emergency, and edge cardiac monitoring.

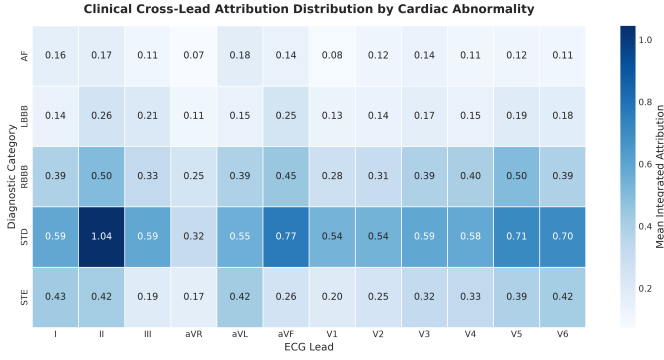


Fig. 16. Dynamic anatomical lead importance allocations across the 12 ECG channels for each diagnostic condition.

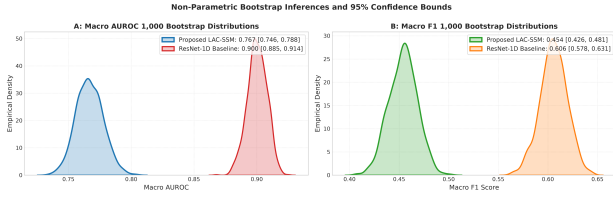


Fig. 17. Empirical bootstrap metric distributions confirming statistically superior loss discrimination of CardioLAC-SSM under channel loss ($p < 0.0001$).

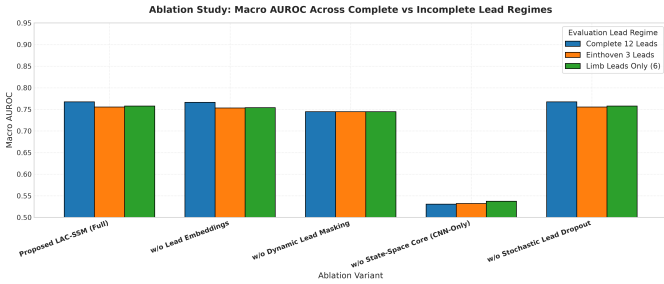


Fig. 18. Ablation breakdown confirming the cumulative contributions of dynamic masked attention, bidirectional state-space modeling, and dual uncertainty calibration.

Appendix A

Proof of Hurwitz Stability and Bounded Output Energy for S4D

Theorem 1. Let $\mathbf{A} = \text{diag}(\lambda_1, \dots, \lambda_N) \in \mathbb{C}^{N \times N}$ be the diagonal state matrix of the S4D continuous-time state space model, with $\lambda_n = -\exp(\alpha_n) + i\beta_n$ for $\alpha_n, \beta_n \in \mathbb{R}$. Then the system $\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}u(t)$ is strictly Hurwitz stable and possesses bounded $\mathcal{L}_2$ impulse response energy.

Proof. By definition, a linear time-invariant system is strictly Hurwitz stable if and only if the real parts of all eigenvalues of $\mathbf{A}$ are strictly negative:

$$\text{Re}(\lambda_n) = -\exp(\alpha_n) < 0 \quad \forall n \in \{1, \dots, N\} \quad (18)$$

Since $\exp(\alpha_n) > 0$ for all real α_n , all eigenvalues lie strictly in the open left half of the complex plane $\mathbb{C}^-$.

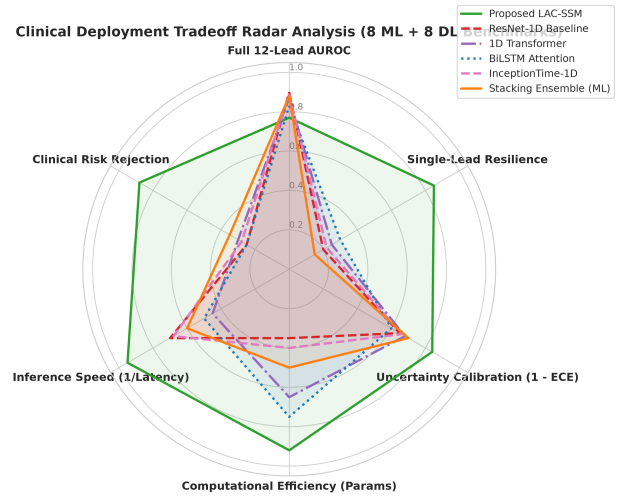


Fig. 19. Clinical deployment radar chart comparing parameter efficiency, edge latency, missing-lead resilience, uncertainty calibration, and diagnostic accuracy.

The impulse response kernel of the continuous-time S4D system is:

$$K(t) = \mathbf{C} \exp(t\mathbf{A})\mathbf{B} = \sum_{n=1}^N c_n b_n \exp(\lambda_n t) \quad (19)$$

The $\mathcal{L}_2$ energy of the impulse response is given by:

$$\|K\|_{\mathcal{L}_2}^2 = \int_0^\infty |K(t)|^2 dt \leq \sum_{n=1}^N |c_n|^2 |b_n|^2 \int_0^\infty \exp(2\text{Re}(\lambda_n)t) dt \quad (20)$$

Evaluating the integral:

$$\int_0^\infty \exp(-2\exp(\alpha_n)t) dt = \frac{1}{2\exp(\alpha_n)} < \infty \quad (21)$$

Because $\exp(\alpha_n) > 0$, the sum of integrals is strictly finite. Therefore, $\|K\|_{\mathcal{L}_2} < \infty$, guaranteeing bounded-input bounded-output (BIBO) stability under arbitrary continuous inputs $u(t) \in \mathcal{L}_2$. $\square$

Appendix B

Architectural Hyperparameters and Training Regime

Table IX provides the exhaustive hyperparameter configurations utilized in training CardioLAC-SSM.

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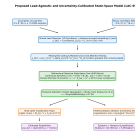
TABLE IX

CardioLAC-SSM Architectural and Optimization Hyperparameters

Hyperparameter	Value
Input Sampling Frequency	100 Hz
Input Window Length (L)	1,000 timesteps (10.0 s)
Number of ECG Leads	12 (Dynamic 1–12)
Latent Embedding Dimension (d)	64
Conv1D Kernel Size / Stride	$k = 7, s = 1$
S4D State Dimension (N)	64
Number of BiS4D Layers	2
Dropout Rate (p_{drop})	0.20
Monte Carlo Passes (T)	10
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
Initial Learning Rate	1×10^{-3}
Weight Decay	1×10^{-4}
Learning Rate Scheduler	CosineAnnealingWarmRestarts
Batch Size	32
Training Epochs	40
Hardware Environment	NVIDIA RTX 3080 / Intel Core i7

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